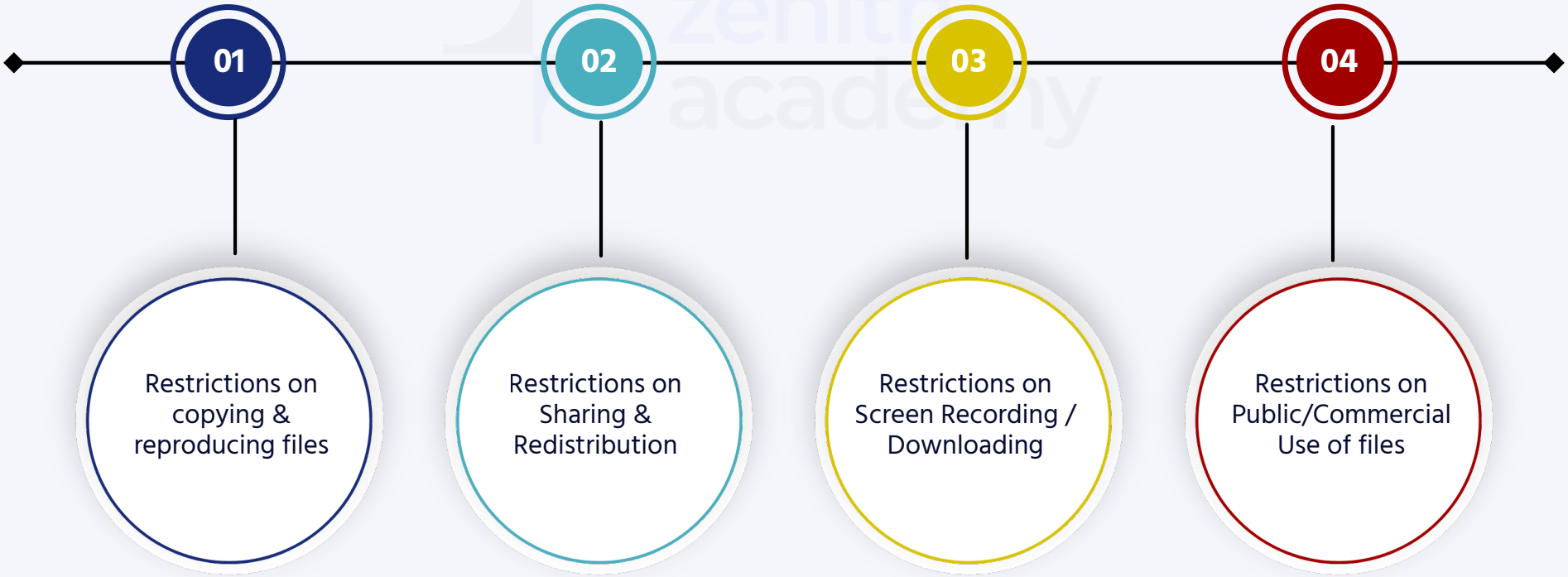
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WATER POLLUTION

Impurities of water

- Dissolved –gases like Nitrogen , carbon dioxide, hydrogen sulphate, ammonia etc.
- Suspended –sand ,mud,slit,clay etc.

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Sources of water pollution

- Sewage
- Industrial and trade waste
- Agricultural pollutants
- Physical pollutants



Indicators of water pollution

- Amount of total suspended solids
- Biochemical oxygen demand (BOD)
- Concentration of chlorides
- Concentration of nitrogen
- Concentration of phosphorus
- Absence of dissolved oxygen

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Water Borne Diseases

A.Biological

B.chemical

A. Biological

1. Due to the presence of an infectious agent
   - Viral- Hep-A,Hep-E ,poliomyelitis, diarrhoea (rota)
   - Bacterial –Typhoid ,Ecoli,cholera etc.
   - Protozoal –Amoebiasis,Giardiasis
   - Helminthic-Round worm, Thread worm , Hydatid disease
   - Leptospiral- Weil’s disease

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1. Due to the presence of an aquatic host
   - Snail-schistosomiasis
   - Cyclops – Guinea worm ,fish tape worm

B . Chemical

- Dental health –fluoride
- Methhaemoglobinaemia- High nitrate content
- Cardiovascular diseases –Hardness of water has beneficial. effects against CVD.

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4. Some diseases are transmitted due to the inadequate use of water
   Eg: Shigellosis,Trachoma,Conjunctivities,Ascariasis,Scabies
5. Some diseases are related to disease carrying insects
   Eg: Malaria,Filaria,Arboviral Infections,African sleeping sickness,etc..

The Water (Prevention and Control of Pollution) Act, 1974

- The Water (Prevention and Control of Pollution) Act, 1974 was enacted to prevent and control water pollution and maintain or restore water health in the country.
- The law was amended in 1988.

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QUESTIONS

1.Percentage of population in India having access to safe water is:
a.50 % b. 39% c.69% d. 90%

2. Which is the wrong statement about safe and wholesome water?
a. Free from pathogens
b. Free from harmful chemicals
c. Free from colour and odour
d. Usable for agriculture purposes

3. Per capita water requirement for all domestic purposes is:
a. 70-80 litres b.80-120 litres c. 120-150 litres d. 150-200 litres

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4. **Purest form of water in nature ?**

a. River water
b. Deep well
c. Rain water
d. Impounding reservoirs

5. Guinea worm disease is a health problem where there are:

a. Tube wells
b. Dug wells
c. Step wells
d. Artesian wells

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6.Safe yield of source of water is defined as the yield that is adequate for:

a. 50% of the year
b. 75% of the year
c. 95% of the year
d. 99% of the year

7. The prime source of all water:

a. Rain water
b. River water
c. Well water
d. Sea water

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8. The term coined for large excavations in which surface water is stored:

a. Well

b. Tank

c. Spring

d. River

9. The safest and the most satisfactory source of water supply:

a Shallow well

b. Deep well

c. Artesian well

d. Tube well

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10. Which of the following is not surface water?

a. River water

b. Ponds

c. Lake

d. Spring

11. Salt concentration in sea water is:

a. 1.5%

b. 2%

c. 3.5%

d. 5%

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12. The cheapest source and most practical means of providing water to small communities:

a. Surface water
b. Ground water
c. Rain water
d. Sea water

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13. Which of the following provides the safest water and often the most satisfactory sources of water supply ?

a. Deep well
b. Springs
c. Arteisan well
d. Shallow well

14.Source of water likely to be free from pathogenic agents:

a. Surface water

b. Ground water

c. Rain water

d. Sea water

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15.Wells which taps water from above the first impervious layer in the ground:

a. Deep well

b. Springs

c. Arteisan well

d. Shallow well

16.Name the place which depends completely on tube wells for water supply

a)Delhi b)Mumbai c) Chandigarh d) Kolkata

17.Shallow tube wells are also known as

a)Bore wells b) Driven wells c) Pucca wells d) Sanitary wells

18.A deep well in which water rises above the ground water level due to pressure between two impervious strata

a. Deep well

b. Springs

c. Arteisan well

d. Shallow well

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19.The distance of a drinking water source from a source of contamination is not less than _____ metres / feet

or

The cone of filtration of a well is:

a)10/25 b)15/50 c) 20/100 d)15/75

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20.The location of a sanitary well is recommended in such a way that no user will have to carry water for more than ____ metre/yards

a. 10

b. 50

c. 75

d. 100

21.The lining of a well should be a depth of atleast____

a. 6 meter

b. 10 meter

c. 5 meter

d. 15 meter

22.The height of the parappet wall of a well should be atleast _____ cms

a. 50-55 cms

b. 55-60 cms

c. 70-75 cms

d. 75-80 cms

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23.A water borne disease caused by snail:

a. Fish tapeworm

b. Guinea worm

c. Schistosomiasis

d. Round worm

24.Groundwater comes to the surface and flows freely under natural pressure is called ____

a. Deep well

b. Springs

c. Arteisan well

d. Shallow well

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25.The water act was passed by the Parliament in the year

a. 1981

b. 1971

c. 1974

d. 1982

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ENVIRONMENTAL SANITATION

2 mark questions

1.What is safe and wholesome water?
2.Explain Impounding reservoir?
3.What is shallow wells and what are it’s disadvantages?
4.Write about tube wells?
5.Which are the indicators of water pollution?

5 mark questions

1.Explain surface water and it’s different classification?
2 .What is wells and explain it’s different kinds.

Purification of water

Purification of water is broadly classified into two types

1. PURIFICATION OF WATER ON A LARGE SCALE

a) Storage

b) Filtration -**slow sand and rapid sand filters**

c) Disinfection- **chlorination, Ozonation,ultraviolet irradiation**

2. PURIFICATION OF WATER ON A SMALL SCALE

a) Household purification

1. PURIFICATION OF WATER ON A LARGE SCALE

- The purpose of water treatment is to produce water, that is safe and wholesome.
- Ground water needs no treatment, but needs disinfection.
- Surface water tends to be turbid and polluted and requires extensive treatment.

It consists of 3 main components

a) Storage

b) Filtration

A) STORAGE

- Water is drawn out from the source and impounded in a artificial reservoir (tank)
- At this time natural purification takes place and 90% of suspended impurities settled down in 24 hrs(physical purification)
- This allows penetration of light into the water.
- Aerobic bacteria oxidizes organic matter in water and free ammonia content is reduced., as a result nitrates appears(chemical purification)
- 90% of the pathogenic organisms killed within 5-7 days(biological purification)
- The optimum period of storage of river water is

B) FILTRATION

- It is the second stage of purification.

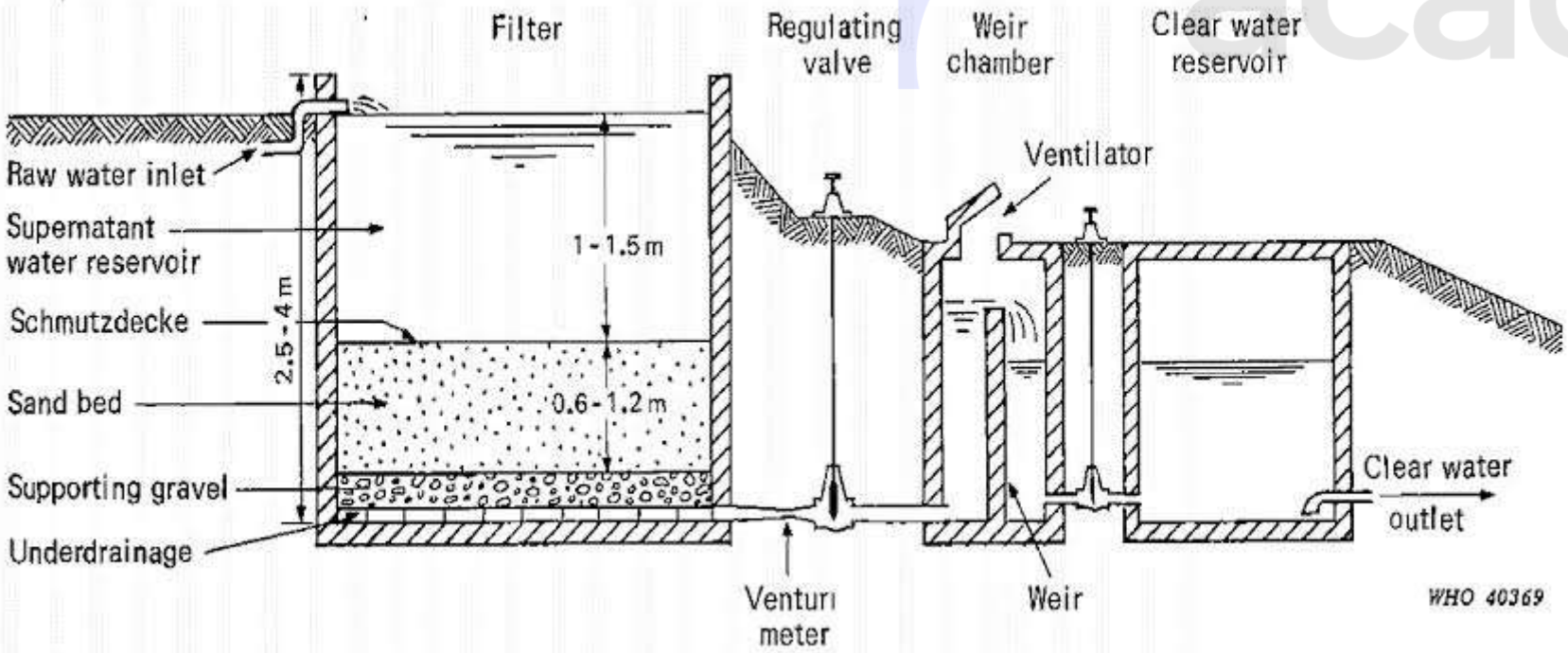
98-99% of bacteria are removed by filtration.

2 types of filters are used for this purpose

I) SLOW SAND FILTERS (BIOLOGICAL FILTERS)

II)RAPID SAND FILTERS (MECHANICAL FILTERS)

I) SLOW SAND FILTERS (BIOLOGICAL FILTERS)



- First used in 1804 in Scotland
- Generally accepted as a standard method of water purification
- Elements of slow sand filters consists of,

a) Raw water

b) Sand bed

c) Under drainage system

d) Filter control Valves

A)Raw water

- Raw water first stored in a settlement tank for 1-2 days
- The depth of raw water above the sand bed is 1-1.5 m.
- During this period natural purification of water (sedimentation, oxidation etc.) takes place.
- 90% of suspended impurities settled down by gravity
- Certain amount of bacteria are also removed.
- Now water becomes clear in appearance.

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B) SAND BED

- Most important part of the slow sand filters.
- Main process is filtration
- The filter bed is an open rectangular box shape , 2.5 m to 4 m deep which consists of 1.5 m of standing water,1.2 m of graded sand and 0.4 m of graded gravel.
- Thickness of sand bed is 1 m.
- The size of sand grains are between 0.2 -0.3 mm.
- The surface area of sand bed is 15,000 m^2 .
- Rate of filtration of water lies between 0.1 and 0.4 m^3/hr per square metre of the sand bed

- When the filter is newly laid, it act as a mechanical filter .
- Very soon the surface of the sand bed is covered with a slimmy growth called vital layer or zoogleal layer or biological layer or schmutzdecke layer.
- This layer consists of plankton, algae , diatoms and bacteria.
- The formation of this layer is called "ripening of the filter" and the vital layer is called the " heart of the slow sand filters "
- This vital layer removes organic matter, holds back bacteria and oxidizes Nitrogen into nitrates, yielding a bacteria free water.

C) UNDER DRAINAGE SYSTEM

- At the bottom of the filter bed ,is the under drainage system.
- It consists of porous pipes
- Provides an outlet for filtered water

D) FILTER CONTROL VALVES

- Main purpose is to maintain a steady rate of filtration
- Venturimeter is an important component of the regulation system which measures the bed resistance or "loss of head"
- When the bed resistance increases, the raw water is drained off and the sand bed is cleaned by 'scrapping' the top portion of the sand layer (vital layer) to a depth of 1-2 cm.
- When the loss of head exceeds 1.3 m , it is uneconomical to run the filter.
- Slow sand filter is usually run for weeks or even months without cleaning.
- After 20-30 scrapping the thickness of the sand bed is reduced to 0.5-0.8 meter , then the

Advantages of slow sand filter

- Simple to construct and operate
- Cost is cheaper
- Physical, chemical and bacteriological quality of filtered water is high.
- Reduce total bacterial count and E.coli count to 99.99%

Disadvantages of slow sand filter

- Sand bed needs reconstruction periodically.